

Ex No. 5 Understand the features of Wireshark and encapsulation of information at various layers at a Protocol stack.

10/8/26

AIM:

Experiments on Packet Capture tool: Wireshark

Packet sniffer:

Sniffs messages being sent / received from / by your Computer.

Capturing and analysing packets using Wireshark tool

To filter, capture, view, packets in Wireshark tool.

Capture 100 packets from the Ethernet.

Procedure:

- Select Local Area Connection in Wireshark.
- Go to Capture option.
- Select stop capture automatically after 100 packets.
- Click start capture.
- Save packets.

1. To display TCP/UDP packets

Procedure:

- Select LA Connection in Wireshark
- Go to Capture option
- Stop capture automatically after 100 packets
- Click start capture.
- Select TCP packets
- To see flow graph.
- Save the packets.

2. To display ARP packets:

- Go to capture option.
- stop capture after 100 packets.
- click start capture.
- search ARP packets in search bar.
- save the packets

3. To display DNS packets:

- Go to capture option
- select stop capture after 100 packets.
- Click start capture.
- search DNS packets in search bar.
- See flow graph click statistics, flow graph.
- save the packets.

4. To display HTTP packets:

- select Local Area Connection in Wireshark.
- Go to capture option.
- select stop capture after 100 packets.
- Click start capture.
- search HTTP packets in search bar.
- save packets.

5. To display IP/ICMP packets:

- Select LA Connection in Wireshark.
- Go to capture option.
- select stop capture after 100 packets.
- click ~~start~~ capture.
- search ICMP / IP packets in search bar.
- save the packets.

6. To display DHCP packets:

- select Local Area Connection in Wireshark.
- Go to capture option.
- select stop capture after 100 packets.
- click start capture.
- search DHCP packets in search bar.
- save the packets.

File Edit View Go Capture Analyze Statistics Sequence Help

Packets

No.	Time	Source	Destination	Protocol	Length	Info
91	2.00000000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.4? Tell 172.16.10.144
92	2.00427000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.156? Tell 172.16.10.144
93	2.00700000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
94	2.00940000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
95	2.01000000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
96	2.01060000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
97	2.01120000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
98	2.01180000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
99	2.01240000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
100	2.01300000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
101	2.01360000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
102	2.01420000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
103	2.01480000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
104	2.01540000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
105	2.01600000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
106	2.01660000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
107	2.01720000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
108	2.01780000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
109	2.01840000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
110	2.01900000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
111	2.01960000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
112	2.02020000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
113	2.02080000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
114	2.02140000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
115	2.02200000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
116	2.02260000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
117	2.02320000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
118	2.02380000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
119	2.02440000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
120	2.02500000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
121	2.02560000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144
122	2.02620000	10.0.0.0.0.0.0	10.0.0.0.0.0.0	ARP	60	Who has 172.16.10.160? Tell 172.16.10.144

[illegible]

Result:

Thus TCP/UDP, ARP, DNS, HTTP, IP/ICMP / DHCP packets was successfully executed.